

Phu Gia Hoang

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SUMMARY

Driven NLP researcher with a strong background in Natural Language Processing, Machine Learning, and AI. My recent research focuses on mechanistic interpretability, particularly developing geometric explanations of the internal representations of large language models. I am interested in understanding LLM representations - how concepts are encoded, organized, and transformed inside models - and how these internal mechanisms shape model behavior. I am passionate about developing interpretable AI systems and studying how humans and machines learn from language, with the goal of improving the fairness, robustness, and reasoning capabilities of AI systems.

EDUCATION

M.Sc., Natural Language Processing

Abu Dhabi, UAE

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

Aug/2023-May/2025

GPA: 3.57/4.0 (Full sponsorship)

Key courseworks: Advanced Natural Language Processing, Speech Processing

Thesis project: Patch-Copying Repeated Tokens for Efficient Generation

B.Sc, Data Science

HCMC, Vietnam

University of Information Technology - Vietnam National University, HCMC

Aug/2019-Apr/2023

GPA: 8.89/10 (Top 2nd of 2023 graduating class)

Key courseworks: NLP, Optimization, ML, DL, Advanced Probability & statistics, Discrete Structure, Linear Algebra, Calculus

Thesis: Hate Speech Spans Detection in Vietnamese (10/10)

RESEARCH EXPERIENCE

Master Thesis MBZUAI

Abu Dhabi, UAE

Academic advisors: Prof. Yova Kementchedjhieva and Prof. Kentaro Inui

Keywords: In-Context Learning, Model Efficiency & Representation Reuse, Information-Theoretic Interpretability

Patch-Copying Repeated Tokens for Efficient Generation

Dec/2024-Present

- Designed a patch-copying mechanism for autoregressive models that reuses token-level internal representations during repeated sequence generation, reducing computational redundancy without modifying model weights and architecture.
- Conducted controlled experiments on synthetic grammatical error correction tasks using Qwen2.5-3B (36-layer model); demonstrated that patching up to 5 layers across repeated spans yields identical outputs compared to unpatched decoding.
- Achieved significant estimated FLOP savings via layer-wise reuse while preserving output quality, confirming a sweet spot between shallow patching and deep representational stability.

The UIT NLP Group, UIT-VNUHCM

HCMC, Vietnam

Academic advisors: Dr. Kiet Nguyen Van. and Prof. Ngan Luu-Thuy Nguyen

Keywords: Hate Speech Detection, Low-Resource Language NLP, Annotation & Dataset Development

Hate Speech Spans Detection in Vietnamese

Mar/2021-Apr/2022

- Contributed to fairness in NLP by developing a novel hate speech detection dataset tailored to low-resource Vietnamese, with annotation schemes reflecting social context..

- Proposed a novel dataset, ViHOS - Vietnamese Hate and Offensive Spans, which consists of 11,056 online comments with advanced annotations on hate speech spans and benchmark experiments.
- Spearheaded training initiatives for 06 annotators which included writing a 16-page annotation guidelines, creating training documents and plans to boost Inner Annotation Agreement (IAA) scores by three annotation-shared phases. These efforts contributed to an increase in the IAA score from 0.71 in the first phase to 0.82 in the final phase.

Hate Speech Detection in Vietnamese

Mar/2021-Jun/2021

- Proposed a two-stage data pre-processing procedure and handed over the development of an efficient combine-model for Vietnamese online comments that outperformed previous studies on ViHSD dataset and HSD-VLSP dataset by 0.0174 and 0.0393 in F1-score, respectively.
- Published in a peer-reviewed journal and presented at a research competition.

PUBLICATIONS

Phong Do, Son Tran, **Phu Hoang**, Kiet Nguyen, and Ngan Nguyen. 2024. VLUE: A New Benchmark and Multi-task Knowledge Transfer Learning for Vietnamese Natural Language Understanding. In Findings of the Association for Computational Linguistics: NAACL 2024, pages 211–222, Mexico City, Mexico. Association for Computational Linguistics. DOI:

<https://doi.org/10.18653/v1/2024.findings-naacl.15>

Phu Gia Hoang, Canh Duc Luu, Khanh Quoc Tran, Kiet Van Nguyen, and Ngan Luu-Thuy Nguyen. ViHOS: Hate Speech Spans Detection for Vietnamese. 2023. In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL-2023), pages 652–669, Dubrovnik, Croatia. Association for Computational Linguistics. DOI:

<https://doi.org/10.18653/v1/2023.eacl-main.47>

Phu Gia Hoang, Le Thanh, and Hai-Long Trieu. 2023. VBD_NLP at SemEval-2023 Task 2: Named Entity Recognition Systems Enhanced by BabelNet and Wikipedia. In Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023), pages 1833–1843, Toronto, Canada. Association for Computational Linguistics. DOI: <https://doi.org/10.18653/v1/2023.semeval-1.253>

Khanh Quoc Tran, An Trong Do, **Phu Gia Hoang**, Canh Duc Luu, Trong Hop Do, and Kiet Van Nguyen. Vietnamese Hate and Offensive Detection using PhoBERT-CNN and Social Media Streaming Data. Neural Computation & Application (2022). DOI: <https://doi.org/10.1007/s00521-022-07745-w>

Phu Gia Hoang, Luan Thanh Nguyen, and Kiet Nguyen. 2021. UIT-E10dot3 at SemEval-2021 Task 5: Toxic Spans Detection with Named Entity Recognition and Question-Answering Approaches. In Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021), pages 919–926, Online. Association for Computational Linguistics. DOI:

<http://dx.doi.org/10.18653/v1/2021.semeval-1.125>

WORKING EXPERIENCE

Research Associate

Darmstadt, Germany

Ubiquitous Knowledge Processing (UKP) Lab, TU Darmstadt

Jun/2025-Present

- Developed geometry-based analysis tools to study how internal representations in LLMs align, organize, and transform across contexts.
- Contributed to the development of a mechanistic interpretability framework that streamlines access to modern interpretability methods, enabling researchers to efficiently analyze and experiment with LLMs.

AI Engineer Internship

Fujairah Research Center

Fujairah, UAE

Jun-Jul/2024

- Collaborated in a multidisciplinary team to develop an AI-powered internal chatbot, improving organizational management.
- Led the design of the chatbot’s user interface and logic flow, optimizing backend operations based on user interactions.
- Applied advanced prompt engineering and developed RAG workflows to integrate cutting-edge retrieval-augmented generation into the chatbot, enhancing accuracy and relevance in responses.
- Contributed to AI-driven problem-solving, with a focus on future research applications in NLP and human-computer interaction.

AI Engineer, Full-time

VinBigData JSC, VinGroup

Hanoi, Vietnam

Jul/2022-May/2023

- Enhanced experience with Python, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing and AI Ethics.
- Authored a paper that describes a system for solving fine-grained NER task in English and Bengal using mDeBERTa with customized self-annotating model, which combines mBERT+Bi-LSTM+CRF and BabelNet. Our augmented system improves of 2.38% and 11.33% over the English and Bangla baselines, respectively in SemEval-2023 Task 2: MultiCoNER II.

HONORS

Student of Five Merits, University of Information Technology - Vietnam National University, HCMC.
Dec/2021

Certificate of merit for being the main author of a paper on a prestigious international magazine of the information science and engineering, recognized by ISI, Director of Vietnam National University, HCMC. Nov/2021

AWARDS

Consolation prize of 23rd "Scientific Research - Euréka", Ho Chi Minh Communist Youth Union and Vietnam National University, Ho Chi Minh City. Dec/2021
Gold Medal of 9th Design, Manufacturing, Application Competition, Ho Chi Minh Communist Youth Union. Dec/2021
Second Prize of 5th Makerthon, Ho Chi Minh Communist Youth Union. Dec/2021
Champion of Entry Category in DIGITAL WATER HACKATHON 2021, Singapore University of Social Science and Ripple2wave. (S\$1,000). June/2021
Champion of AI APP HACKATHON 2020, Anscenter, DeepHub, and Magik Coworking Space. Oct/2020

SKILLS

Programming: Python, Pytorch, Transformers, XGBoost, NumPy, Pandas, Optuna, Pyspark.
Research and Analytical Skills: Research methodologies, data analysis, academic writing.
Others: Visualization (Matplotlib, Seaborn), effective communication, collaboration.